MIL-L-51336
5 November 1969
(See section 6)

MILITARY SPECIFICATION

LEAD OXIDE, RED, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of red lead oxide powder, Pb_3O_4 (hereinafter referred to as lead oxide).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" X 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179".)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

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3. REQUIREMENTS

3.1 Chemical characteristics. Lead oxide shall conform to the chemical characterisitics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay as Pb_3O_4	98.0	----	4.2.4.1
Insoluble matter in nitric acid	----	0.10	4.2.4.2
Water-soluble substances	----	0.05	4.2.4.3
Carbon compounds (as C)	----	0.010	4.2.4.4
Manganese (Mn)	----	0.0005	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the lead oxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof, shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Fiberboard packing aids not as specified.
(f) Tape or strapping not as specified.
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Assay as Pb_3O_4. Weigh to the nearest milligram (mg) approximately 1 gram (g) of the specimen into an Erlenmeyer flask. Add 30 milliliters (ml) of 1 to 1 nitric acid and then add 20.0 ml of dilute hydrogen peroxide (1 volume of 30 percent hydrogen peroxide to 30 volumes of water). Stir until no more specimen dissolves, add 25 ml of water, and titrate the excess hydrogen peroxide with 0.1N potassium permanganate solution. Titrate also with 0.1N potassium permanganate solution a blank containing 20.0 ml of the same diluted hydrogen peroxide to which has been added 30 ml of 1 to 1 nitric acid and 25 ml of water. Calculate the percent lead oxide as Pb_3O_4 as follows:

$$\text{Percent lead oxide} = \frac{34.28\ A\ (B - C)}{W}$$

where: A = Normality of the potassium permanganate solution,
B = Milliliters of potassium permanganate solution required to titrate the blank,
C = Milliliters of potassium permanganate solution required to titrate the specimen, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter in nitric acid. Weigh to the nearest milligram approximately 2 g of the specimen. Add 40 ml of water and 3 ml of concentrated nitric acid, then add with stirring, sufficient 30 percent hydrogen peroxide to dissolve the specimen. Filter through a tared filter crucible, wash with water, dry to constant weight at 105° ± 5° C, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter in nitric acid as follows:

$$\text{Percent insoluble matter in nitric acid} = \frac{100\ (A - B)}{W}$$

where: A = Weight of filter crucible and residue in grams,
B = Weight of filter crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Water-soluble substances. Weigh to the nearest milligram approximately 10 g of the specimen and boil with 250 ml of water for 5 minutes. Cool to room temperature and filter. Evaporate the filtrate to dryness, gently ignite, and weigh to the nearest 0.1 mg. Calculate the percent water-soluble substances as follows:

$$\text{Percent water-soluble substances} = \frac{100\ A}{W}$$

where: A = Weight of filtrate residue in grams and
W = Weight of specimen in grams.

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4.2.4.4 Carbon compounds. Weigh 2.000 g of the specimen and heat in a current of carbon dioxide-free air or oxygen at red heat. Pass the evolved gases through a mixture of 30 ml of water and 0.5 ml of 28-percent ammonium hydroxide solution. Add 5 ml of 12-percent barium chloride solution. Any turbidity produced shall be no greater than that produced when 1.8 mg of anhydrous sodium carbonate is tested in the same manner as the specimen.

4.2.4.5 Manganese. Weigh 5.000 g of the specimen, decompose with 15 ml of concentrated hydrochloric acid, then add 10 ml of concentrated sulfuric acid, and evaporate to copious acid fumes. Cool to room temperature, cautiously add 20 ml of water and 0.5 g of lead dioxide, and heat on a steam bath for 5 minutes. Dilute to 50 ml with water and filter through asbestos. Any color produced in the filtrate shall be no darker than that produced when 0.025 mg of manganese ion is treated in the same manner as the specimen except for the omission of the 15 ml of concentrated hydrochloric acid.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. One pound ± 0.1 ounce of lead oxide shall be packaged in a type I, class 1 glass bottle with a plastic or metal screw cap, in inclosure b, as specified in PPP-C-301. The bottle shall also conform to group A, class 1, type optional, style 2, light penetration grade 2, closure A or B, with outer seal A of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of lead oxide, packaged as specified in 5.1.1, shall be intermediately packaged as specified in PPP-C-301 except that only one tier containing three rows of four bottles per row shall be placed in each intermediate container.

5.1.2 Level B. Packaging and intermediate packaging shall be as specified in 5.1.1 and 5.1.1.1 except that inclosure b of PPP-C-301 shall not be required.

5.1.3 Level C. Packaging and intermediate packaging shall be as specified in 5.1.1 and 5.1.1.1 except that inclosure b of PPP-C-301 shall not be required and the intermediate packaging box shall conform to style RSC, type CF, class domestic, variety SW, grade 175 of PPP-B-636. The box shall be closed as specified by method I or II in the appendix to PPP-B-636.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages of lead oxide shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box. The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated plywood box shall conform to type overseas, style I or J, for a type 2 load of PPP-B-601. The wirebound wood box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. Fiberboard fillers and pads shall be used in accordance with PPP-C-301 to assure a tight pack. Each container shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. Unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for no less than 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Packing shall be as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636 and no preservative shall be required. Each box shall be reinforced by pressure-sensitive adhesive filament reinforced tape as specified in the appendix to PPP-B-636.

5.2.3 Level C. Packing shall be as specified in 5.2.1 except that the box shall conform to style RSC, type CF, class domestic, variety optional, grade 275 of PPP-B-636 and no preservative shall be required. The box shall be closed as specified by method I or II in the appendix to PPP-B-636.

5.3 Marking. In addition to the marking specified in 5.3.1, unit, intermediate, and shipping containers shall be marked in accordance with Department of Transportation regulations and MIL-STD-129.

5.3.1 Special marking. Each bottle of lead oxide shall be durably and legibly labeled to show the following information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Net weight of contents.
(e) Assay and maximum limits of impurities.
(f) Precautionary marking:

WARNING: HARMFUL DUST

Avoid breathing dust.
Wash thoroughly before eating or smoking.
Keep away from feed or food products.
Keep away from children.

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5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using load type I except that strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness. The load tie-down straps parallel to the 48-inch pallet dimension shall be modified to serve as load unitizing straps, containing the load, but not the pallet. One load tie-down strap shall be added to, and centered between, those straps which are normally required to be parallel to the 40-inch pallet width. One horizontal strap of 0.75-inch minimum width and 0.023-inch minimum thickness shall be required for each tier of containers. The pallet shall conform to style 1, type optional, class 1, preserved of MIL-P-15011. Any wood storage aids, if used, shall be preserved as specified for the pallet.

6. NOTES

6.1 Intended use. Lead oxide is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-164, dated 10 October 1952.

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Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0575

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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